DIPLOMA PROGRAMME PRACTICE · SL · SET B

Chemistry / Paper 1A

Question booklet · 30 marks · 90 minutes across 1A and 1B

Answer every question. Choose one option, A–D, per question. Use a calculator and a clean Chemistry data booklet. Record choices online or on separate paper.

QUESTION 1 [1]

Crystals of iodine form on a cold surface above warm iodine. In a second vessel a white solid forms where gaseous HCl meets NH_3. Which classification is correct?

A Both changes are chemical because solids form

B Both deposits result from sublimation

C The iodine undergoes a physical change; the acid and base form a new substance

D The iodine change is chemical and NH_4Cl formation is physical

QUESTION 2 [1]

An ion has mass number 52, 28 neutrons and 21 electrons. Which change gives a neutral atom of the same isotope?

A Lose seven neutrons

B Gain three protons

C Lose three electrons

D Gain three electrons

QUESTION 3 [1]

Two samples contain only isotopes of relative masses 20 and 22. Equal numbers of atoms from samples of mean masses 20.2 and 21.0 are combined. What percentage of atoms in the mixture have mass 22?

A 20%

B 40%

C 30%

D 60%

QUESTION 4 [1]

Which species has the greatest number of unpaired electrons in its gaseous ground state? Use atomic numbers N 7, O 8 and F 9.

A O

B F

C F^-

D N

QUESTION 5 [1]

A laboratory needs 0.0120 mol of chloride ions in 150.0 cm³. What volume of 0.200 mol dm⁻³ $CaCl_2$ stock should be diluted to that volume?

A 30.0 cm³

B 90.0 cm³

C 15.0 cm³

D 60.0 cm³

QUESTION 6 [1]

At the same temperature and pressure, 0.300 g of gas A occupies 150 cm³ and 0.400 g of gas B occupies 100 cm³. Both are ideal. What is M(B)/M(A)?

A 2.00

B 1.33

C 0.750

D 0.500

QUESTION 7 [1]

A rigid vessel at 280 K contains 0.600 mol of ideal gas at 84.0 kPa. After gas escapes, pressure is 81.0 kPa at 360 K. How much gas remains?

A 0.800 mol

B 0.450 mol

C 0.150 mol

D 0.579 mol

QUESTION 8 [1]

A solid containing only Mg^{2+} and N^{3-} shatters when layers are displaced. Which pair gives its formula and the cause of fracture?

A MgN; a sea of electrons becomes immobile

B Mg_3N_2; neutral molecules separate

C Mg_3N_2; like charges are brought together

D Mg_2N_3; unlike charges repel

QUESTION 9 [1]

Why can ethanol and methoxymethane both accept hydrogen bonds from water, while only one forms conventional hydrogen bonds with itself?

A Ethanol must ionize before interacting with water

B Both have O lone pairs; only ethanol has O–H

C Both have O–H; only the ether has lone pairs

D Only methoxymethane contains polar bonds

QUESTION 10 [1]

Which molecule has polar bonds whose dipoles cancel because of its molecular geometry?

A SO_2
B CH_3F
C BF_3
D NF_3

QUESTION 11 [1]

An alloy conducts in both the solid and molten states. Which observation most directly distinguishes metallic conduction in the solid from ionic conduction in a molten salt?

A The sample warms during current flow
B Charge moves while the metal ion cores remain on lattice sites
C The material has a high density
D Its surface is shiny

QUESTION 12 [1]

The repeat of poly(propene) is C_3H_6. Ignore end groups. A chain has relative mass 42,000, and C=12, H=1. How many backbone carbon atoms does it contain?

A 2000
B 6000
C 1000
D 3000

QUESTION 13 [1]

A period-3 element forms a basic oxide X_2O. Which comparison with the element immediately to its right is correct?

A X has the same nuclear charge
B X has a larger atomic radius and fewer valence electrons
C X has a smaller atomic radius and fewer valence electrons
D X has a larger atomic radius and more valence electrons

QUESTION 14 [1]

In the reaction $Cl_2+2Br^-\rightarrow 2Cl^-+Br_2$, what happens to the reducing agent?

A It remains as chloride ions
B Its oxidation state decreases from 0 to −1
C Its oxidation state increases from −1 to 0
D It accepts two electrons per atom

QUESTION 15 [1]

A compound has formula C_4H_8O and a terminal –CHO group. Which name is consistent with both facts?

A Methyl propanoate
B Butan-2-one
C Butan-1-ol
D 2-methylpropanal

QUESTION 16 [1]

A sample labelled 'butan-2-ol' is converted to a ketone without carbon–carbon bond cleavage. Which statement about the starting material and product is correct?

A The product is a carboxylic acid
B The alcohol gains H_2 and gives butanal
C Both have molecular formula $C_4H_{10}O$
D The alcohol loses two H atoms overall and its four-carbon skeleton is retained

QUESTION 17 [1]

A burner transfers heat to 150 g of water. Its temperature rises by 8.0 K. Use c=4.18 J g^{-1} K^{-1}. The burner loses 0.400 g of fuel. What is the measured energy transferred per gram burned?

A 50.2 kJ g^{-1}
B 5.02 kJ g^{-1}
C 12.5 kJ g^{-1}
D 2.01 kJ g^{-1}

QUESTION 18 [1]

Estimate ΔH for $H_2(g)+Cl_2(g)\rightarrow 2HCl(g)$ using bond enthalpies H–H 436, Cl–Cl 243 and H–Cl 431 kJ mol^{-1}.

A +183 kJ mol^{-1}
B −183 kJ mol^{-1}
C +248 kJ mol^{-1}
D −431 kJ mol^{-1}

QUESTION 19 [1]

Fuel P releases 50 MJ kg^{-1} and makes 2.5 kg CO_2 per kg. Fuel Q releases 30 MJ kg^{-1} and makes 1.8 kg CO_2 per kg. At equal efficiency, which statement is correct?

A Both give 0.050 kg CO_2 per MJ
B Q releases less CO_2 per MJ
C A comparison is impossible without molar masses
D P releases less CO_2 per MJ

QUESTION 20 [1]

At constant T and p, 60 cm³ CO is mixed with 40 cm³ O_2 and ignited. After cooling to the starting temperature, what total gas volume remains? Assume complete reaction.

A 70 cm³
B 100 cm³
C 60 cm³
D 80 cm³

QUESTION 21 [1]

A synthesis converts one mole each of A and B to one mole each of C and D. Molar masses in g mol^{-1} are A 60, B 46, C 88 and D 18. What is atom economy for desired product C, and does an 80% isolated yield change it?

A 83.0%; no
B 88.0%; no
C 80.0%; yes
D 66.4%; yes

QUESTION 22 [1]

A tablet contains 0.840 g $NaHCO_3$ (M=84.0) and an inert binder. What volume of 0.250 mol dm^{-3} HCl reacts exactly with the bicarbonate?

A 40.0 cm³
B 80.0 cm³
C 20.0 cm³
D 10.0 cm³

QUESTION 23 [1]

A student compares 0.10 g and 0.20 g of a catalyst using equal excess substrate. Which measurement best tests the effect of catalyst mass on initial rate?

A Compare tangents at time zero on product-volume curves under the same conditions
B Compare only the total duration including setup
C Compare final temperatures using different initial temperatures
D Compare final gas volumes after complete reaction

QUESTION 24 [1]

A sealed equilibrium A(g) ⇌ 2B(g) has a forward exothermic reaction. Which single change increases the equilibrium amount of B?

A Add a catalyst at fixed temperature
B Expand the container at fixed temperature
C Remove A at fixed volume and temperature
D Increase the temperature at fixed volume

QUESTION 25 [1]

A reversible reaction in a sealed flask reaches equilibrium. An isotopic label introduced in trace quantity later appears in both reactants and products. What does this establish?

A Reactant and product concentrations are equal
B The equilibrium composition is changing continuously
C The equilibrium constant is one
D Forward and reverse reactions continue at equilibrium

QUESTION 26 [1]

Two monoprotic acids both neutralize 25.0 cm^3 of the same NaOH when equal volumes are used. Acid P has the higher initial pH. What follows if both tests use pure aqueous solutions at the same temperature?

A P necessarily has the smaller concentration
B Equal analytical concentrations; P is less extensively ionized
C P has more ionizable protons per molecule
D Equal concentrations and equal acid strength

QUESTION 27 [1]

Mix 20.0 cm^3 of 0.100 mol dm^{-3} HCl with 10.0 cm^3 of 0.100 mol dm^{-3} NaOH. Assuming additive volumes, what is the pH?

A 2.00
B 1.48
C 1.00
D 7.00

QUESTION 28 [1]

In a Zn/Zn^{2+} and Ag^{+}/Ag galvanic cell, Zn is oxidized. For every 0.020 mol Zn lost, which change occurs?

A 0.010 mol Ag deposits
B 0.020 mol Ag dissolves
C 0.040 mol Ag deposits
D 0.040 mol Ag dissolves

QUESTION 29 [1]

During substitution of methane with bromine, which pair of elementary steps gives the net propagation reaction $CH_4+Br_2\rightarrow CH_3Br+HBr$?

A $Br\bullet+CH_4\rightarrow HBr+CH_3\bullet$ and $CH_3\bullet+Br\bullet\rightarrow CH_3Br$
B $CH_3\bullet+CH_3\bullet\rightarrow C_2H_6$ and $Br_2\rightarrow 2Br\bullet$
C $Br_2\rightarrow 2Br\bullet$ and $2Br\bullet\rightarrow Br_2$
D $Br\bullet+CH_4\rightarrow HBr+CH_3\bullet$ and $CH_3\bullet+Br_2\rightarrow CH_3Br+Br\bullet$

QUESTION 30 **[1]**

Which observation most directly supports the presence of a C=C bond in a colourless hydrocarbon under ordinary test conditions?

A Bromine water is decolourized in the dark

B It burns to produce CO_2

C It forms a colourless gas on boiling

D It is insoluble in water